

Chapter 1

INTRODUCTION

1.1 Background and Context

1.1.1 Overview of Sentiment Analysis in Mobile Applications

Sentiment analysis is a worth exploring topic in natural language processing where the target is to classify the opinion it expresses using a given piece of text. Among different types of dataset, app store reviews completely suit this task because people write them voluntarily, in large numbers, and without being prompted by a survey. One study carried out by Pang et al. (2002) revealed that supervised learning works for sentiment classification although their experiments assumed balanced classes distribution. This is something you can hardly see in practice.

Mobile apps have made user reviews far more abundant. Some popular apps such as Google Play and the Apple App Store have millions of them, and many contain opinions about specific features. However, most tools only care about wheather a review is positive or negative, which ignores useful information. Aspect-Based Sentiment Analysis (ABSA) takes a different approach as it identifies the particular product dimensions a user mentions and assigns sentiment to each one individually (Tang et al., 2016). Product teams, therefore, tend to find this more actionable than aggregate scores.

There are many methods that have been introduced recently for this task. Fine-tuning pre-trained transformer models, BERT for example, for aspect extraction and sentiment classification has outperformed earlier neural methods. The reason is that these models produce context-aware representations, which are better at capturing how an aspect term relates to the sentiment around it (Xu et al., 2019). Most ABSA research, however, targets English. Vietnamese and many other languages still lack comparable datasets and benchmarks.

App store reviews also present challenges that general-purpose models were not designed for. These challenges can be: Users mix slang, abbreviations, and sometimes switch between languages mid-sentence. Moreover, Multilingual models can handle some of this, but they tend to dilute language-specific patterns. Nguyen and Nguyen (2020) argue that pre-trained models built on actual Vietnamese text work better than multilingual alternatives for exactly this reason.

1.1.2 Vietnamese Fintech NLP Challenges

Vietnamese poses real problems for NLP tools when the language is monosyllabic, word boundaries are often unclear, and its tonal system means that small phonetic shifts carry different meanings. Standard NLP toolkits, however, were designed around popular languages in the world and handle none of this well (Vu et al., 2018).

Texts commented by users also bring about difficulties because younger Vietnamese write in "teen code," a set of informal phonetic substitutions and abbreviations, which cannot be seen often in famous datasets. They also use emoji, borrow English words, and switch between Vietnamese and English within a single sentence. Tokenization and normalization pipelines were not built for this kind of input, so the text that reaches the model is often noisy, which results in lower performance.

In fintech, the problem is hard to ignore. Momo, Vietnam's largest digital wallet, passed 30 million registered users in 2024 (VnExpress, 2024), generating hundreds of thousands of daily transactions and a steady stream of user reviews. Most of those reviews are complaints, so reading them all by hand is not realistic, and the Vietnamese NLP tools and datasets that do exist were not designed around fintech-specific language.

PhoBERT, a transformer-based model pre-trained on Vietnamese text, improved performance on these tasks. This model achieved stronger results on various tasks such as part-of-speech tagging, named entity recognition, and text classification benchmarks. Its Vietnamese-specific training makes it a reasonable candidate for analyzing informal, domain-specific reviews. But whether it actually beats multilingual models in a real fintech setting is still an open question, and that is what this study tests.

1.2 Problem Statement

1.2.1 Limitations of Existing Sentiment Analysis Approaches

Traditional approach may not work well for app reviews. Support Vector Machines and Naive Bayes classifiers score reasonably on balanced benchmarks, but their accuracy drops when class distributions are not guaranteed (Medhat et al., 2014). And they usually are balanced where unhappy users' reviews are found far more often than satisfied ones.

Classification based on documents assigns one label to an entire review, but a single review might praise an app's interface and complain about transaction failures and unclear fees in the same sentence. The detail that product teams actually need when deciding what to fix (Maalej & Nabil, 2015) may be thrown away when collapsing that into "positive" or "negative". Vietnamese NLP research has mostly focused on the document level and has not addressed aspect-level sentiment closely.

On top of that, LSTM and GRU architectures model sequences better than traditional methods, but they still have their own weakness with imbalanced data. If negative reviews make up 80% or more of the dataset in total, these models may learn to produce "negative" label almost every time. This is dangerous as overall accuracy still looks acceptable, but the model misses positive and neutral reviews entirely.

On top of that, the standard remedies for class imbalance (SMOTE, random undersampling, class-weighted loss functions, and various combinations) work differently depending on the model.

Specifically, an oversampling approach that improves a Transformer's recall can degrade a recurrent model's precision while the Vietnamese ABSA literature has largely not tested these interactions, so there is little guidance on which combinations actually work for which architectures.

1.2.2 Gap in Vietnamese Fintech ABSA Research

Vietnam's digital payment sector has gained popularity, but not so many people have studied aspect-based sentiment analysis for Vietnamese fintech apps. Vietnamese NLP research has picked up since PhoBERT and VnCoreNLP were published, however publicly available ABSA datasets for Vietnamese are still scarce and need more attention. None of them cover fintech, which has its own vocabulary around transactions, fees, promotions, and support interactions.

The lack of suitable tools has real costs. Fintech companies either sample reviews manually or rely on aggregate sentiment scores that cannot tell them whether users are angry about transaction failures or about slow customer support. That two problems require very different approaches, but document-level analysis labels them together as "negative."

In this study's dataset, 81.3% of reviews are negative with the average review is 101 characters long. At the same time, Users write in slang, abbreviate freely, and switch between Vietnamese and English mid-sentence, making languages processing even more challenging. Existing models and class-imbalance strategies were not designed for all of these at once, and to fill this gap, this study contributes an annotated fintech review dataset and a systematic comparison of models evaluated under these realistic, imbalanced conditions.

1.3 Research Objectives

1.3.1 Dataset Construction and Aspect Annotation

This study builds and publicly releases a Vietnamese fintech ABSA dataset collected from real app store reviews. We collect 4,325 Google Play reviews over 286 days and label each for sentiment and six service aspects: Transaction, UI/UX, Security, Customer Support, Fee, and Promotion.

There is no available Vietnamese fintech dataset with aspect-level annotations published before. The collection window was long enough to capture shifts in what users complain about over time, and the six aspects cover the core service areas of a digital wallet product.

Each review received both a sentiment label and one or more aspect tags from the fintech taxonomy. This is because sentiment and aspects are annotated separately, the dataset supports evaluating each task on its own or training them jointly through multi-task learning.

1.3.2 Comparative Model Evaluation Under Imbalanced Conditions

The second goal is to compare modern Transformer-based models on Vietnamese fintech ABSA under heavy class imbalance. They are LSTM, BiLSTM, GRU, LSTM with attention, PhoBERT, and XLM-RoBERTa. Each of them is tested with four imbalance-handling strategies, namely no balancing, class-weighted loss, SMOTE oversampling, and a hybrid that combines undersampling with SMOTE.

The comparison tends to focus on which combination of model and strategy performs reliably across all three sentiment classes using different metrics. Firstly, Macro-F1 is the primary metric because it

treats every class equally and a model that scores high accuracy by predicting "negative" almost every time will still get a low Macro-F1. Secondly, Weighted-F1, Balanced Accuracy, and class-level precision and recall are reported as well to give a more insights.

This evaluation also tests a specific question still remaining: does pre-training on Vietnamese text give PhoBERT a measurable advantage over XLM-RoBERTa for informal, domain-specific reviews? The two models share the same architecture but were pre-trained on different data, so the comparison should isolates the effect of language-specific training.

1.4 Research Questions

1.4.1 Does Vietnamese-Specific Pre-Training Outperform Multilingual Scale?

The first thing that need addressing is that a model pre-trained specifically on Vietnamese outperform a multilingual model trained on far more data or not. We all know that PhoBERT and XLM-RoBERTa share the same RoBERTa architecture, therefore comparing them isolates the effect of language-specific versus large-scale multilingual pre-training. Language-specific pre-training has shown an edge on standard Vietnamese NLP benchmarks (Pires et al., 2019), however, those benchmarks use relatively clean, formal text.

The outcome has implications for other languages in a similar position. If PhoBERT consistently outperforms XLM-RoBERTa on this kind of noisy, domain-specific data, that strengthens the case for building dedicated monolingual models for underrepresented languages with growing digital economies. But if multilingual models perform comparably once paired with the right class-imbalance strategy, they offer a faster path to deployment, since they skip the cost of collecting and pre-training on monolingual corpora.

1.4.2 How Do Imbalance-Handling Strategies Interact with Model Architectures?

The second question is how class imbalance strategies interact with model capacity, which are all designed to improve minority class detection, but they do not necessarily help every model the same way. A reweighting scheme that works for a large pre-trained Transformer might destabilize a smaller recurrent network, especially when the minority class is very small to begin with.

Practitioners tend to choose an imbalance strategy and a model architecture as if the two choices were independent. A strategy that improves PhoBERT's recall on positive reviews could degrade an LSTM's precision on the same class, so this study tests those interactions directly, then finally the results offer concrete guidance on which combinations work and which ones to avoid when building sentiment classifiers on skewed review data.

1.5 Contributions

1.5.1 Dataset and Annotation Framework

This study's first contribution is the dataset itself with 4,325 Momo digital wallet reviews from Google Play, each annotated for sentiment and tagged with one or more of six service aspects. The dataset is published at <https://www.kaggle.com/datasets/viteanh/momo-reviews>.

The annotation framework is a contribution on its own where the six aspects (Transaction, UI/UX, Security, Customer Support, Fee, and Promotion) were selected based on what digital wallet users actually talk about in app store reviews. In addition, the taxonomy is narrow enough to capture fintech-specific concerns but portable enough to apply to other Vietnamese digital wallets like ZaloPay or VNPay, extending the utility of the framework beyond the immediate dataset.

1.5.2 Experimental Findings

PhoBERT achieves the highest Macro-F1 at 0.5886, compared to 0.5353 for BiLSTM and 0.4912 for XLM-RoBERTa, and Vietnamese-specific pre-training outperforms multilingual scale on this kind of informal fintech text. Prior work showed a similar advantage on cleaner Vietnamese NLP benchmarks, these results suggest the gap holds, and possibly widens, on noisier data.

No single imbalance-handling strategy works reliably across all six models. On the one hand, Class-weighted loss collapses LSTM and GRU to near-zero Macro-F1 but gives Transformers small gains under certain conditions. Additionally, SMOTE helps Transformers slightly and does little for recurrent models. On the other hand, the hybrid approach improves PhoBERT's Balanced Accuracy at the cost of overall accuracy, trading aggregate correctness for better minority class recall. The takeaway is that imbalance strategy and model architecture cannot be chosen independently. Multi-task learning does not improve sentiment classification performance over single-task training for either Transformer model. However, XLM-RoBERTa achieves substantially higher aspect detection F1 than PhoBERT under the multi-task setting, suggesting that multilingual representations transfer more effectively across related tasks despite weaker sentiment classification performance.

1.6 Scope and Limitations

This study focuses only on Momo reviews collected from the Google Play Store, so the findings may not generalize to other Vietnamese fintech applications, different app marketplaces, or digital payment services in other countries. In addition, the dataset covers a single 286-day period, which may not capture changes in user sentiment over longer time spans.

The dataset is highly imbalanced, with 81.3% negative reviews, making it difficult to evaluate minority classes reliably, while the neutral class consistently achieves the lowest performance across all models. In addition, most reviews are very short, with a median length of only 72 characters, providing limited context and causing ambiguous or single-word comments, such as "ok", to be misclassified. The models also struggle with informal language, including slang, teen code, emojis, code-switching, sarcasm, implicit negation, and mixed-language expressions, indicating that standard pre-training and fine-tuning are insufficient for these linguistic patterns. Finally, inter-annotator

agreement was not measured, so future studies should include agreement metrics to improve the reliability and consistency of the annotations.

1.7 Thesis Structure

The thesis has five chapters.

Chapter 1 defines the three problems: why sentiment analysis on Vietnamese fintech reviews is hard to handle, what the goal is, and where the boundaries of the work lie.

Chapter 2 surveys the literature on sentiment analysis in fintech , the change from attention-based recurrent models to Transformer fine-tuning in ABSA, the particular challenges of Vietnamese NLP, and class imbalance methods.

Chapter 3 is about the methodology, which introduce a complete pipeline.

Chapter 4 reports results: overall model performance across each imbalance strategy, class-level and aspect-level metrics, single-task versus multi-task comparisons, training dynamics, and error analysis on the kinds of Vietnamese text that give models the most trouble.

Chapter 5 summarizes the findings, discusses what they mean for Vietnamese NLP and fintech practice, and also covers the study's limitations, and outlines future work, such as expanding the dataset to other Vietnamese wallets and testing instruction-tuned language models.